Investing in Primary Care: The Case of Brazil's Family Health Program

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Abstract

In this study, I investigate whether the expansion of healthcare investments in Brazil, implemented through the Family Health Program (FHP), improved maternal and infant health outcomes. I employ a regression discontinuity design by exploring discontinuities in funding that created a quasi-experimental assignment of FHP resources to municipalities. Using administrative data from various sources from 2004 to 2012, I obtain information on program implementation, health outcomes, local-level facilities, and indicators of access. I provide evidence that FHP investments had significant effects on maternal and infant health outcomes over this period. Results show that a 70% increase in FHP annual transfers led to a 9 percentage point rise in FHP coverage and substantial improvements in birth outcomes from 2004 to 2012. Specifically, birth weight increased by 35 to 41 grams, and there was an approximate 1.8 percentage point increase in full-term births in treated municipalities. Prenatal care also improved significantly, with an increase of about 6 prenatal visits per 100 women of reproductive age. However, I found no significant effects on maternal and infant mortality rates during this time frame. Overall, this paper contributes to understanding the impact of primary healthcare investments on maternal and infant health, opening possibilities for future research.

Key Words: Primary health care, maternal mortality, infant mortality, birth outcomes

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1 Introduction

Over the past decades, numerous countries have implemented initiatives to extend health coverage to poorer populations, providing a rich context for examining the effects of healthcare investments on public health outcomes. These efforts span both developed and developing nations, encompassing a wide range of approaches from subsidized health insurance schemes to the establishment of universal healthcare systems. Brazil's experience has provided a particularly compelling context for examining the impact of such initiatives, especially in the realm of primary healthcare.

In 1988, when the new constitution was enacted, half of Brazil's population had no health coverage (Jurberg and Humphreys, 2010). Since then, the country has undergone substantial changes building the Unified Health System (*Sistema Único de Saúde, SUS*) and now is considered to be one of the most successful cases of health care system reform in developing countries (Harris, 2014). Public policies in healthcare aimed to overcome the flaws of the former system, such as insufficient coverage in some regions of the country and a limited network of primary care centers. One of the most renowned strategies that has been implemented to meet this challenge is the Brazilian Family Health Program (*Programa Saúde da Família*), a nationally scaled model of basic care services within underserved communities. FHP was a milestone in the shift from a centralized model structured around public hospitals in main urban areas to a decentralized one, where the first point of contact between the population and the public health system was placed in local communities (Rocha and Soares, 2010). Besides providing medical care for previously uncovered families, the program helped to lessen the pressure on public hospitals, which would then focus resources to address complex medical conditions.

In this paper, I study the impact of FHP transfers on maternal and infant health outcomes. More specifically, by exploring a discontinuous rule for program funding introduced in 2004, I investigate whether municipalities that received more FHP resources experienced improvements in maternal and infant health outcomes, the program's focus. To achieve this objective, I use administrative records from the Brazilian Ministry of Health's System of Information (DataSUS) and several sources to obtain information on program implementation, health outcomes, local-level facilities, and indicators of access. This dataset, which includes individual-level data on outpatient care, mortality, and birth records, enables me to assess the impact of this new funding rule from 2004 to 2012.

Results show that FHP transfers had significant effects on maternal and infant health from 2004 to 2012. The new funding policy consistently increased FHP coverage, with a 70%

increase in FHP annual transfers leading to a 9 percentage point rise in coverage approximately. It also boosted the number of FHP teams per capita and significantly increased home visits. Prenatal care saw substantial improvements, with an increase of about 6 prenatal visits per 100 women of reproductive age. These findings suggest that the new funding policy effectively enhanced the FHP's capacity to provide maternal care and improved access to primary healthcare services.

The policy led to improvements in some maternal and infant health outcomes. Birth weight consistently increased across all years, with a 70% increase in FHP transfers resulting in an additional 35 to 41 grams of birth weight. Full-term births (gestation of at least 37 weeks) increased by 1.34 to 1.77 percentage points from 2004 to 2010. Maternal hospitalization rates increased, particularly for conditions not sensitive to primary care, suggesting improved access to necessary hospital care. However, the study found no significant effects on maternal and infant mortality rates, possibly due to the rarity of these events and the limited time window of the study. These findings underscore the complexity of health interventions and the importance of targeted funding in achieving specific health outcomes.

The findings of this study both align with and extend the existing literature on primary healthcare interventions in developing countries. The observed increases in indicators of access, such as FHP coverage and prenatal care visits, are consistent with previous research in Brazil, including Rocha and Soares (2010) and Bhalotra and Rocha (2019). The substantial increases in birth weight (35-41 grams) are notably larger than effects found in some comparable studies and is particularly notable given that Bhalotra and Rocha (2019), studying FHP's expansion from 1996 to 2004, found no significant changes in birth weight or other indicators of birth quality. This study's significant effect on birth weight could be attributed to the specific nature of the FHP and its targeted funding increase after 2004. Our observed reductions in low birth weight incidence (up to 1.90 percentage points) also appear to be more substantial than those found in some previous studies. The increase in full-term births aligns with literature showing positive impacts of primary care on gestational age. The lack of significant effects on mortality rates, despite improvements in intermediate outcomes, is consistent with other short to medium-term studies in this field, highlighting the need for longer exposure periods to observe changes in these ultimate health outcomes.

Overall, this study contributes to the literature in several important ways. First, it specifically examines the effects of the 2004 funding rule change, providing insights into how financial incentives can drive program expansion and health outcomes. Crucially, it demonstrates that even a relatively modest increase in funding led to significant improvements in

health outcomes, suggesting that targeted investments can yield substantial benefits. This finding is particularly important for resource-constrained settings. Second, the comprehensive analysis of various outcomes, including birth weight, full-term births, and maternal hospitalizations, offers a more comprehensive view of FHP's impact. By examining intermediate outcomes like prenatal visits and home visits, this study sheds light on the pathways through which FHP might improve health outcomes. Additionally, by analyzing data over an eight-year period, this study offers insights into the program's sustained impact over time, an aspect not extensively covered in previous literature. The focus on birth weight and full-term births adds to our understanding of FHP's impact on important indicators of infant health that have long-term consequences. Finally, the finding of increased maternal hospitalizations for non-primary care sensitive conditions suggests improved access to necessary hospital care, an aspect not extensively covered in previous literature. These nuanced findings highlight the complexity of health interventions and the need for further research on factors influencing ultimate health outcomes like mortality rates.

The paper proceeds as follows. I explain the historical background and context of the Family Health Program in Section 2. Section 3 describes the data. In Section 4, I discuss the empirical strategy. Section 5 presents the results, and Section 6 concludes.

2 Institutional Background

2.1 Primary Health Care in Brazil

Over recent decades, Brazil has made significant progress in health outcomes, exceeding the Millennium Development Goals targets. There has been a substantial decrease in mortality rates, particularly for infants and children, with infant mortality dropping to 12.59 per 1,000. Life expectancy has increased remarkably, reaching 76.4 years at birth. For comparison, in 1991, the infant mortality rate was 45.1 per 1,000, and life expectancy was 65 years (DATASUS, 2023). Many studies attribute these improvements to the extensive expansion of primary health facilities across the country, demonstrating the critical role of this expansion in enhancing health outcomes (Macinko, Guanais and Marinho De Souza, 2006; Aquino, de Oliveira and Barreto, 2009; Gragnolati, Lindelow and Couttolenc, 2013; Guanais, 2013).

The 1988 Federal Constitution marked a significant reform in Brazil's health system. For the first time, the state became legally responsible for ensuring free and universal health care. To achieve this, the Unified Health System (Sistema Único de Saúde, SUS) was imple-

mented to provide universal care at no cost to patients, decentralizing health provision to municipal governments.

Before the introduction of *SUS* in the 1990s, public health in Brazil relied on a corporate structure administered by the Ministry of Health and the social security system. Certain occupational categories had access to adequate medical care through retirement and pension institutes. However, most of the population had to rely on inadequate public services or pay out-of-pocket for private healthcare services (Paim, Travassos, Almeida, Bahia and Macinko, 2011).

In 1994, the Family Health Program (FHP) was introduced as part of the government's policy to extend primary care access. Following pilot programs in several municipalities with community-based primary care, health coverage was expanded to cover over 85 million people by 2006, reaching more than 90% of municipalities (Brazilian Ministry of Health, 2006*a*). Concurrently, federal expenditure on the program increased from R$ 280 million in 1998 to R$ 2.7 billion in 2005 (Bhalotra and Rocha, 2019). The expansion of the FHP is displayed in Figure 12.

The program operates through family healthcare teams, each consisting of one family doctor, one nurse, one assistant nurse, and four to six health community agents. Since 2004, some expanded teams have also included a dentist, an assistant dentist, and a dental hygiene technician. Each team serves 600–1000 families from a specific neighborhood or community, providing care both in primary health units [1] and in households. FHP teams serve as the initial point of contact between families and the health system, offering primary healthcare, health counseling, and preventive and recovery services (Rocha and Soares, 2010; Paim et al., 2011; Bhalotra and Rocha, 2019).

According to the Brazilian Ministry of Health, we can summarize FHP's main characteristics as follows: i) to serve as an entry point into a hierarchical and regional health system; ii) to have a defined territory and population under the responsibility of a specific health team, establishing liability (co-responsibility for the health care of a defined population); iii) to intervene in critical risk factors at the community level; iv) to perform comprehensive, continuous, and quality care; v) to promote education and health awareness activities; vi) to encourage community organization and act as a link between different sectors of civil society; and vii) to use information systems to monitor decisions and health outcomes (DAB, 2000).

The program has proven to be an effective model, improving medical conditions at a low

[1] Operational basis of a FHP team in a specific geographical area

cost, primarily through prevention and early detection. By regularly monitoring the same families, FHP professionals can promote better dietary and hygiene habits, detect early symptoms, and minimize the effects of endemic problems. Moreover, this community-level approach allows most basic conditions to be treated within the program itself, reducing pressure on public hospitals. (Bhalotra and Rocha, 2019) show that the increase in outpatient care under the FHP reduced hospital caseloads, allowing hospitals to accommodate more inpatient procedures.

In 2000, the annual cost of maintaining one family health team was estimated to be between R$215,000 and R$340,000, or US$109,610 to US$173,400. Assuming that a health team covers approximately 3,500 individuals, this corresponds to a yearly cost of US$31 to US$50 per person covered (FGV-EPOS, 2001). Administratively and financially, the FHP is a federal program implemented in a decentralized manner by municipalities. While implementation should involve coordination across municipal, state, and federal governments, there are instances of implementation without state government involvement. FHP implementation requires voluntary participation of local administrations, preferably with state support. The responsibilities of different government levels are defined as follows ((Brazilian Ministry of Health, 2006*b*), as translated by (Rocha and Soares, 2010)):

- Federal Government: Develop national basic health policy goals; co-finance the "basic health attention" system; organize human resource training; propose mechanisms to program, control, regulate, and evaluate the "basic health attention" system; maintain the national database;
- State Government: Monitor FHP implementation and execution; regulate inter-municipal relations; coordinate human resource qualification policies; co-finance the program; assist in executing basic health attention system strategies; and
- Municipal Government: Define and implement the FHP model; hire program staff; maintain the basic health unit management network; co-finance the program; maintain the information system; evaluate basic health attention teams under its supervision.

In 2004, the government established new rules for FHP transfers to municipalities. Municipalities with a Human Development Index (HDI) below 0.70 and a population under 30,000 began receiving 50% additional funding per team. Previously, FHP transfers were paid monthly as a fixed amount per team for all municipalities (approximately R$ 7,130 per team per month, plus R$ 20,000 in implementation incentives per team). Under the new rule, qualifying municipalities could elect to receive around R$ 10,695 per team per

month. [2] The list of municipalities eligible for extra funding released in 2004 remained unchanged even after the publication of new population estimates and the 2010 census. The HDI for eligibility was calculated based on the 2000 census, and the population was referenced using the 2003 estimates from the government's statistics department (IBGE). [3] By examining frequency breaks around the cutoffs and applying tests developed by (McCrary, 2008), we find no evidence of manipulation at eligibility cutoffs. Therefore, we have strong theoretical and empirical reasons to believe that local political authorities could not have manipulated eligibility for treatment.

Extensive empirical evidence exists regarding the program's impact on health outcomes. (Macinko et al., 2006) document a statistically significant effect of the FHP on infant mortality using state-level data (27 states). By surveying subjective perceptions, (Macinko and De Sá, 2007) report that the program's presence in a municipality is associated with better-perceived health among the population. (Rocha and Soares, 2010) use municipality-level data to conduct a comprehensive analysis of FHP effects on mortality by age group, cause of death, region, and initial mortality level. The results show a robust correlation between program implementation timing and mortality reductions across the age distribution, particularly for infant mortality. The program's impact is larger in municipalities with worse initial conditions and in Brazil's poorest regions (North and Northeast). Using a similar empirical strategy, (Bhalotra and Rocha, 2019) further examine the mechanisms through which the FHP impacted mortality outcomes, demonstrating that the program improved access to both primary and hospital care, especially for conditions less treatable with primary care.

3 Data

The following sections describe my main datasets and I present some descriptive statistics on Table 1.

3.1 Family Health Program Data

I obtain data related to the implementation of the FHP at the municipality level from the Brazilian Ministry of Health (Department of Basic Attention, MS/DAB). Updated monthly, it provides the number of FHP teams and the proportion of the population covered in each municipality.

[2] The population limit was set to 50,000 for states that form the legal Amazon

[3] The original list is constant of the following decree: Portaria 1.434/GM, July 14, 2004.

3.2 Health Data

Brazilian Ministry of Health (MS/Datasus) also provides data on health outcomes and access to health care. The National System of Information on Hospitalizations (Datasus/SIH) contains administrative information at the hospitalization level, comprehending all hospital admissions funded by the Brazilian Unified Health System. It provides information on cause of hospitalization, duration of stay, final outcome (discharge or death), socioeconomic characteristics of the patient (municipality and zipcode of residence, gender, and date of birth), costs, level of complexity and type of bed (regular or ITU). The data are managed by the Health Care Agency (SAS/Ministry of Health) with support of local and regional public health agencies, which receive information about hospitalizations from public and private hospitals through standardized inpatient forms, AIHs (*Autorização de Internação Hospitalar*). The National System of Information on Ambulatory Care (Datasus/SIA) includes administrative information on ambulatory visits funded by SUS. All medical care provided on an outpatient basis, including diagnosis, observation, consultation, treatment, intervention, and rehabilitation services are included. We are also able to identify the type of health professional that delivered the service (physicians by specialization, nurses, or community health agents). Ambulatory visits may take place, for instance, in clinics, hospitals, health facilities that provide low-complexity primary health services, and PSF units. Information on hospital infrastructure (number of hospital beds and presence of hospital in the municipality) is also obtained from the Ministry of Health (MS/Datasus).

3.3 Other sources

Finally, we employ other municipality data that are auxiliary to our analysis. The National Treasury database (FINBRA) provides data on basic health transfers to municipalities. Annual data on the municipality population, by age and gender, are obtained from the Brazilian Census Bureau (IBGE, after Instituto Brasileiro de Geografia e Estatística). All variables are collapsed at the municipality-by-year level, and merged with the other data containing FHP variables.

4 Empirical Strategy

Consider the following cross-sectional relationship between FHP transfers and maternal-infant health outcomes:

$$y_i = \beta_0 + \beta_1 log(T_i) + x_i'\lambda + \varepsilon_i$$

where y_i measures maternal and infant health outcomes in municipality i (e.g., infant mortality rate, low birth weight rate), T_i is the value of federal resources from the FHP, x_i is a vector of observed municipal characteristics, and ε_i and ν_i are unobserved determinants of health outcomes and FHP funds, respectively. Under the assumption that $E[\varepsilon_i \nu_i] = 0$, the least squares estimator of β_1 will be a consistent estimate of the causal effect of FHP transfers on maternal and infant health outcomes.

However, there are several potential omitted variables that covary with both FHP transfers and health outcomes. For instance, if FHP resources are intentionally allocated to areas with poorer health indicators, then estimates of health improvements could be biased downward (or upward). Another concern arises if specific areas, such as poorer regions, are more likely to benefit from public spending choices and also exhibit distinct health trends that are unrelated to FHP investments.

To overcome these empirical concerns, we exploit an exogenous variation in federal transfers to municipal governments from the Family Health Program in Brazil to identify how additional resources in primary healthcare impacts maternal and infant health outcomes. As described in Section 2, funding changed discontinuously according to population and HDI thresholds. Here, a municipality m with population p_m, and HDI denoted by h_m, has the ATE defined over the frontier:

$$F = (p_m, h_m) : \left(p_m <= \overline{P}, h_m = 0.7\right) \cup \left(p_m = \overline{P}, h_m <= 0.7\right)$$

with respective treatment cutoffs for population and HDI at $\overline{P}$ and 0.7. [4]

We employ a fuzzy regression discontinuity design (RDD) in an effort to generate an assignment of FHP investment that is as good as random near the cutoffs, allowing causal inference of its effects. Intuitively, municipalities just below the threshold should be, on average, similar in all observed and unobserved characteristics to those just above it, so

[4]The population threshold $\overline{P}$ is set at 30,000 for most states. However, for municipalities in states that form the Legal Amazon, the population threshold is set at 50,000. This difference in thresholds is accounted for in our analysis through normalization of the running variable.

that any difference in outcomes between these two groups must be caused by variations in FHP transfers. We take advantage of this discontinuity to evaluate the effects of higher FHP transfers on maternal and infant health outcomes.

Given that FHP funding across municipalities is a discontinuous function of both municipal population and HDI, quasi-random assignment of this variable could be achieved by employing multivariate regression-discontinuity design (MRDD), where multiple assignment variables and cutoffs may be used for treatment assignment. In the two-score case, the average treatment effect is identified for a frontier of points, in contrast to a single point in the one-score case. This changes the estimation and interpretation of the treatment effects within the RD framework, mainly due to the potential heterogeneity of these effects along the frontier.

In our RD setting, however, exploring the population-threshold is more appealing because the subsamples have a much larger number of observations (see Figure 1). Therefore, we employ a fuzzy RDD with one running variable in an effort to generate an assignment of FHP investment that is as good as random near the cutoffs, allowing causal inference of its effects.

We estimate variations of the following regression discontinuity design for each year from 2004 to 2012:

$$y_{it} = \beta_0 + \beta_1 \hat{T}_{it} + f(P_{it}) + X'_{it}\gamma + \varepsilon_{it}$$

$$T_{it} = \alpha_0 + \alpha_1 1\{P_{it} < \overline{P}\} + g(P_{it}) + X'_{it}\delta + \nu_{it}$$

where P_{it} is the normalized population of municipality i in year t, 1. is an indicator function that equals one if the municipality's normalized population is below the population cutoff $\overline{P}k$, and X_{it} is a vector of municipal controls. The functions $f(.)$ and $g(.)$ are local linear functions of population.

One potential pitfall of using discontinuity-based assignment is that local officials might strategically manipulate indicators to obtain desirable policies. Manipulative sorting could then introduce selection bias into estimates of causal effects and invalidate the comparability of municipalities near the threshold.

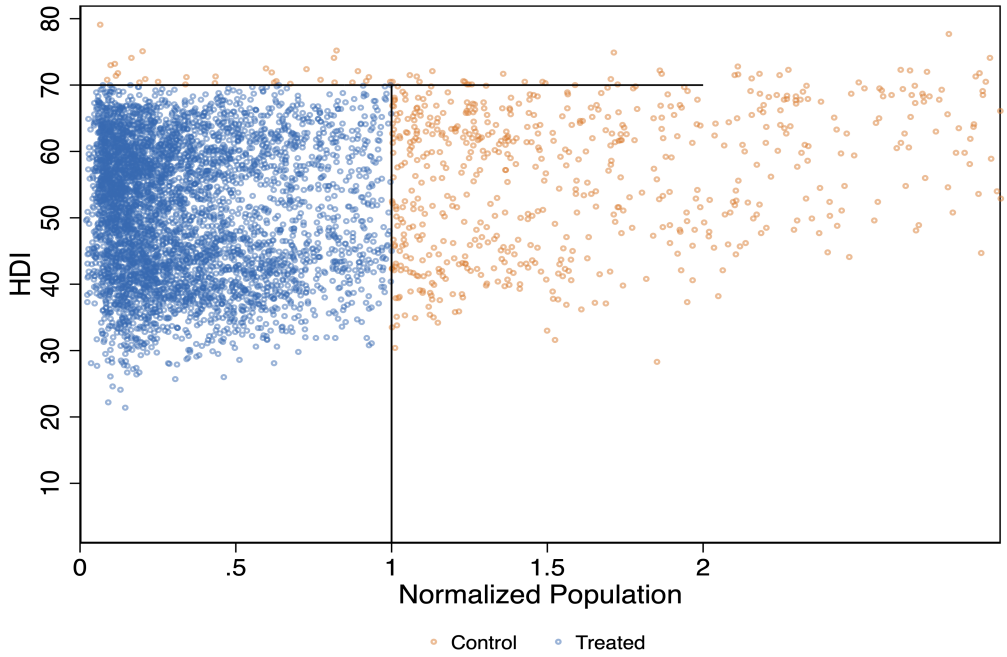
In section A.1, we present some robustness checks we used to validate our empirical strategy. There is no evidence of manipulation in the running variables or discontinuities in observable characteristics around the thresholds. By investigating frequency breaks around the cutoffs and applying the tests developed by McCrary (2008), we find no evidence of

manipulation at eligibility cutoffs. As an initial assessment of how serious the issue of manipulative sorting might be, we plot the population histogram (Figure 13), supporting that there is no significant jump in the distribution just below the threshold. Not surprisingly, the McCrary test does not indicate frequency breaks around the cutoffs (Figure 14). We fail to reject the null hypothesis of no manipulation at the threshold. This consistent pattern across all years provides strong evidence against systematic manipulation of the population figures around the cutoff point. The lack of significant discontinuity in the density of the running variable at the threshold supports the validity of the regression discontinuity design in this context, suggesting that the assignment of municipalities to treatment and control groups can be considered as good as random in the neighborhood of the cutoff.

All this is consistent with the fact that the list of municipalities eligible for the extra funding released in 2004 did not change even after the publication of new population estimates and a new census in 2010. The HDI for eligibility was calculated based on the 2000 census, and the population was referenced using the 2003 estimates of the government's statistics department (IBGE). [5] Therefore, we have good theoretical and empirical reasons to believe that local political authorities could not have manipulated the eligibility for treatment.

[5]The original list is constant of the following decree: Portaria 1.434/GM, July 14, 2004.

Figure 1: Potential Sample and Treatment Frontier with Normalized Population



Source: Brazilian Ministry of Health, SAS/Dept de Atenção Básica – DAB. The original source of data used to compute each variable is listed on Table 2
a

[a]This figure shows the distribution of municipalities across the treatment frontier. The x-axis represents the normalized population, where 1 is the cutoff (30,000 for regular states, 50,000 for Legal Amazon states). The y-axis shows the Human Development Index (HDI). Blue points represent treated municipalities (normalized population < 1 and HDI < 70), while orange points are control municipalities. The vertical line at x = 1 represents the population cutoff, and the horizontal line at y = 70 represents the HDI cutoff. Legal Amazon includes municipalities in the North region, Maranhão (MA), and Mato Grosso (MT).

5 Results

My empirical strategy explores the population eligibility criteria as a source of exogenous variation in FHP transfers to study its effects on maternal and infant health outcomes using an instrumental variables framework. The causal estimation depends on the existence of a strong first-stage relationship. In Figure 15, I start with a visual inspection to check if FHP transfers to municipalities were sensitive to the population threshold across the years 2004-2012. Crossing the population threshold set up by these rules is highly and significantly predictive of federal FHP transfers to municipalities, as shown by each graph. A consistent and clear discontinuity at the population threshold is evident throughout all years, demonstrating the persistent effectiveness of the FHP funding rule in directing more resources to municipalities just below the cutoff. Notably, the magnitude of this discontinuity appears to have increased over time, with the gap in FHP expenditures between municipalities on either side of the threshold widening considerably from 2004 to 2012.

Table 4 presents the first-stage estimates of the impact of the population threshold on FHP transfers from 2004 to 2012. The results show a consistently significant effect across all years, with coefficients ranging from 0.4798 to 0.7849. The control group mean (average transfers for municipalities just above the threshold) also increased substantially, from R$270,879.19 in 2004 to R$734,016.87 in 2012. This overall increase in funding, combined with the growing treatment effect, suggests that the FHP funding policy may have been refined or its implementation strengthened over time, leading to more pronounced differences between eligible and non-eligible municipalities. This substantial and sustained increase in funding provides context for the observed effects on various health indicators and underscores the importance of considering both short-term and long-term impacts of the policy.

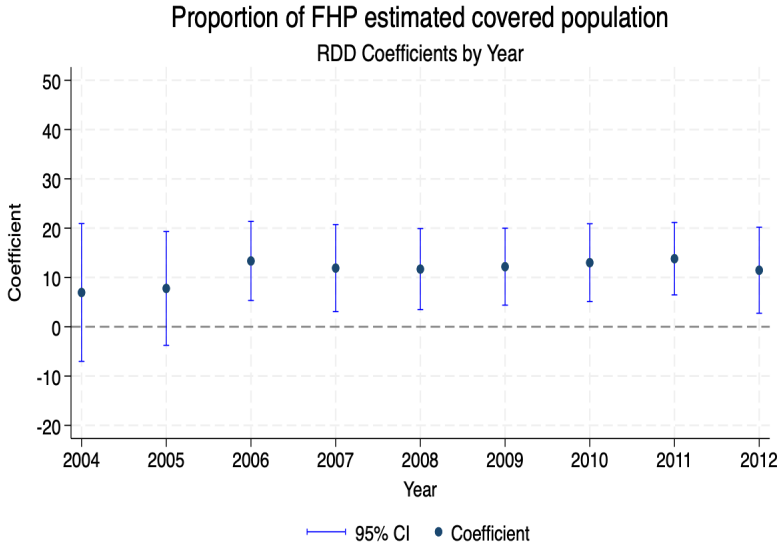
5.1 Access to health services

First, I investigate indicators of access to primary care, focusing most of the analysis on maternal and infant health, the priorities of the program. Unless otherwise noted, all estimations include municipal controls, such as *Bolsa Família* Program (PBF) [6] coverage, GDP per capita (log), urban population, and percentage of households with basic sanitation services, piped water, and sewer. I rely on the Calonico et al. (2014) algorithm for choosing optimal bandwidths, but also explore alternative choices of the bandwidth.

[6] *Programa Bolsa Família* is a conditional cash transfer program implemented in Brazil in 2003, aimed at reducing poverty and inequality by providing financial assistance to low-income families.

Figure 2 shows the RDD coefficients for FHP estimated covered population from 2004 to 2012. Results suggests that after an initial implementation period, the new funding policy substantially and consistently boosted FHP coverage in eligible municipalities with consistent positive effects from 2006 onward. A 70% increase in FHP annual transfers resulted in an approximately 9 percentage point increase in covered population. Given control group coverage ranged from 65.1% (2004) to 83.8% (2012), this increase is substantial. This suggests effective use of additional FHP resources to expand coverage in eligible municipalities. Detailed results are in Table 5 in the appendix.

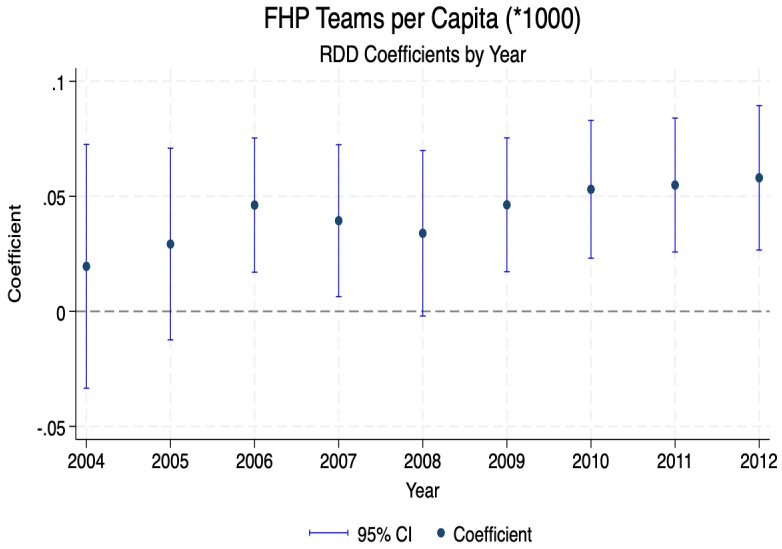
Figure 2



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on FHP covered population from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

I also analyse the effects on FHP Teams per 1000 population from 2004 to 2012. In Figure 3, we can see that after an initial adjustment period, consistent positive effects emerged from 2006 onwards. A 70% increase in FHP annual transfers led to a rise of about 0.028 to 0.042 teams per 1000 population. This effect is substantial when compared to the control group, which grew from 0.147 per 1000 in 2004 to 1.265 per 1000 in 2012. For instance, in 2006, with a control group average of 0.438 teams per 1000, the treatment effect of 0.0323 represented a 7.4% relative increase. Detailed results are available in Table 6 in the appendix.

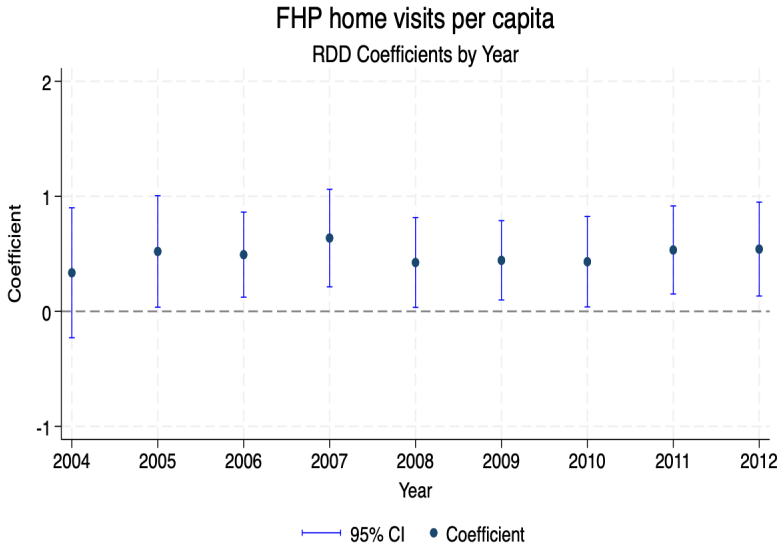
Figure 3



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on FHP Teams per Capita (*1000) from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

The impact of the new FHP funding policy on home visits per capita is depicted in Figure 4. Starting from 2005, the policy consistently led to a significant increase in home visits. After 2007, the effect stabilized around 0.4 additional visits per capita, remaining statistically significant. This sustained rise in home visits suggests that the new funding mechanism effectively enhanced the FHP's outreach capabilities, potentially improving access to primary care for harder-to-reach populations. Comprehensive results are available in Table 7 of the appendix.

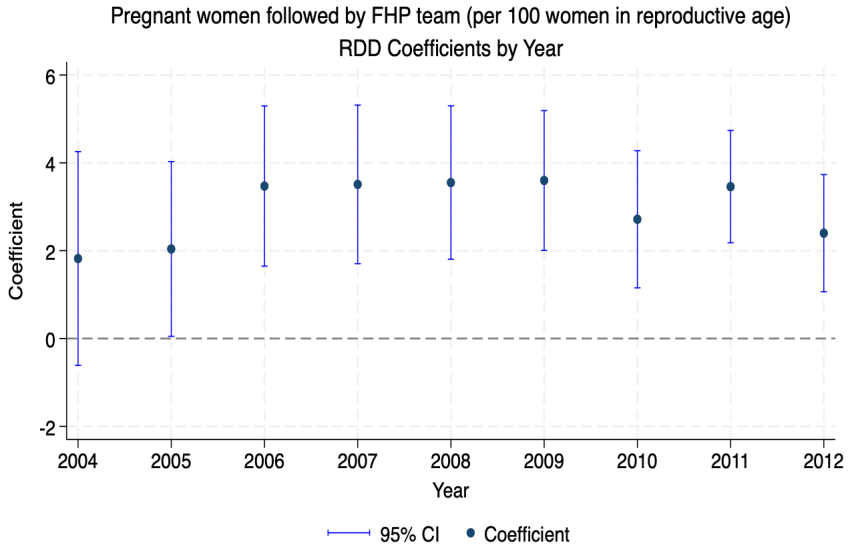
Figure 4



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on FHP home visits per capita from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

Figure 5 showcases the RDD coefficients for pregnant women followed by FHP teams per 100 women of reproductive age from 2004 to 2012. The magnitude of this impact is considerable, with coefficients around 2-3 additional pregnant women followed per 100 women of reproductive age for a 70% increase in FHP transfers. To put these figures into perspective, the control group (municipalities just above the population threshold) averaged between 15.837 and 18.906 pregnant women followed per 100 women of reproductive age annually. This consistent and significant improvement suggests that the new funding mechanism effectively enhanced the FHP's capacity to support maternal health, potentially contributing to better pregnancy outcomes. Full details of this analysis are available in Table 8 in the appendix.

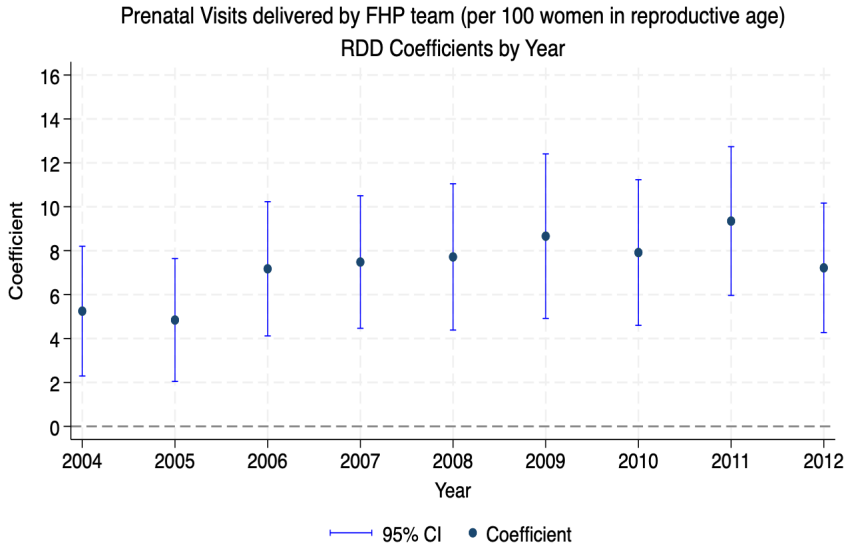
Figure 5



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on pregnant women followed by FHP team (per 100 women in reproductive age) from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

Figure 6 illustrates the impact of the new FHP funding policy on prenatal visits delivered by FHP teams per 100 women of reproductive age from 2004 to 2012. The results reveal a consistently significant and positive effect across all years. The policy's effect was particularly strong in later years, reaching its peak in 2011 with an increase around 6.6 visits for a 70% increase in transfers, representing a 33.5% boost over the control group's mean of 19.991. This sustained and substantial rise in prenatal visits suggests that the new funding mechanism significantly enhanced the FHP's capacity to provide maternal care, potentially leading to improved pregnancy outcomes and maternal health. Detailed results can be found in Table 9 of the appendix.

Figure 6



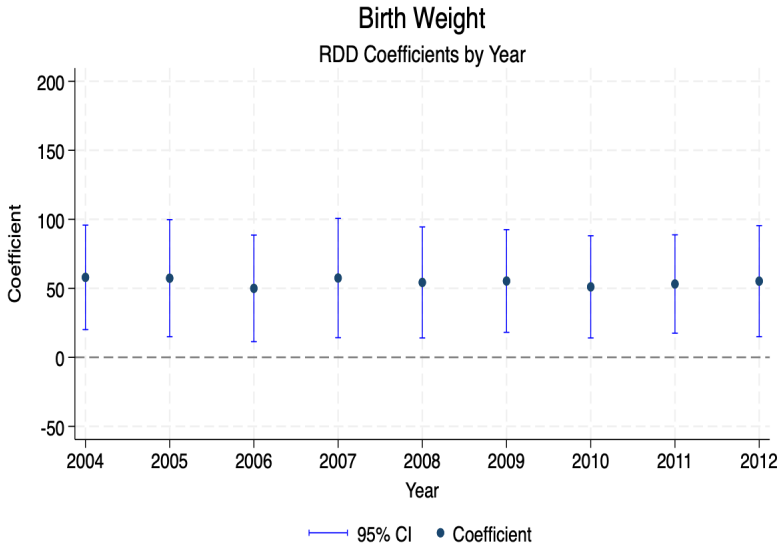
Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on prenatal visits delivered by FHP team (per 100 women in reproductive age) from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

5.2 Maternal and Infant Health Outcomes

Next, I present the estimates for maternal and infant health outcomes over the period 2004-2012. All estimations include municipal controls, such as PBF coverage, GDP per capita (log), urban population, and percentage of households with basic sanitation services, piped water, and sewer. I rely on the Calonico, Cattaneo and Titiunik (2014) algorithm for choosing optimal bandwidths, but also explore alternative choices of the bandwidth. To help evaluate the magnitude of the effects, I present the dependent variable mean (average for municipalities within a 7000-population interval above the population-threshold).

Figure 7 displays the RDD coefficients for birth weight (in grams) from 2004 to 2012, revealing the impact of the new FHP funding policy on this crucial health indicator. The results show a consistently positive and statistically significant effect across all years. The magnitude of this impact is noteworthy, with coefficients ranging from approximately 35 to 40 grams of additional birth weight for a 70% increase in FHP transfers. This persistent positive impact on birth weight suggests that the enhanced FHP funding led to improved prenatal care and maternal health services, potentially contributing to better long-term health outcomes for newborns. Comprehensive results are available in Table 10 of the appendix.

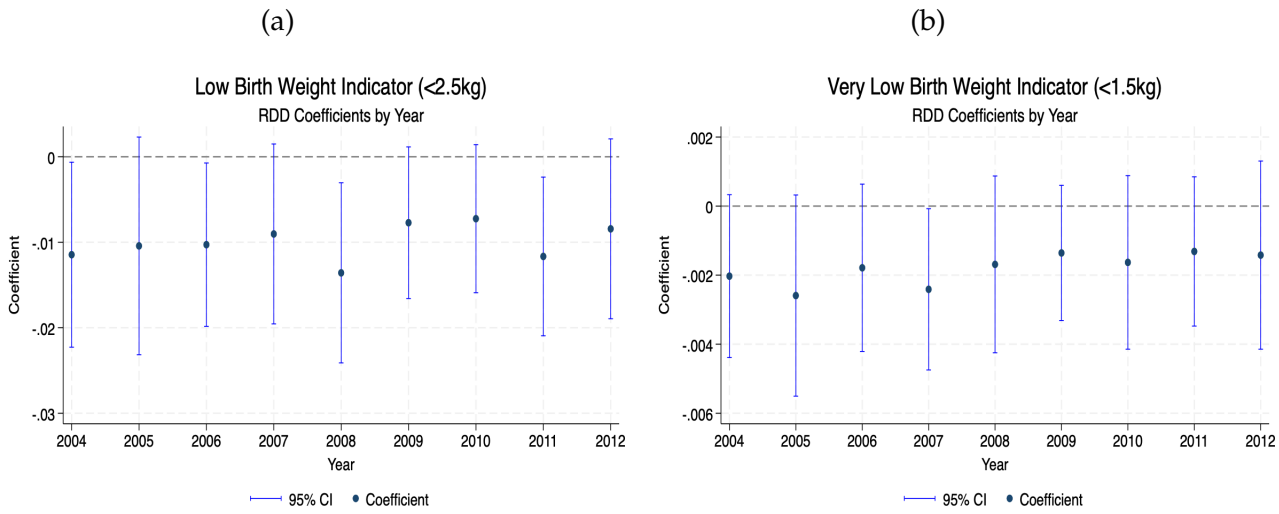
Figure 7



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on birth weight (in grams) from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

Figures 8a and 8b present the RDD coefficients for low birth weight (<2.5kg) and very low birth weight (<1.5kg) indicators from 2004 to 2012. For low birth weight, the effects are consistently negative, suggesting a potential reduction in the incidence of low birth weight babies. The very low birth weight indicator shows a similar trend, with smaller magnitudes and less statistical significance. The absence of more detectable effects, particularly for very low birth weight, may be attributed to factors such as the rarity of these events. While these findings hint at potential improvements in birth outcomes due to enhanced FHP funding, the inconsistent statistical significance across years suggests caution in interpretation. Detailed results are available in Tables 11 and 12 in the appendix.

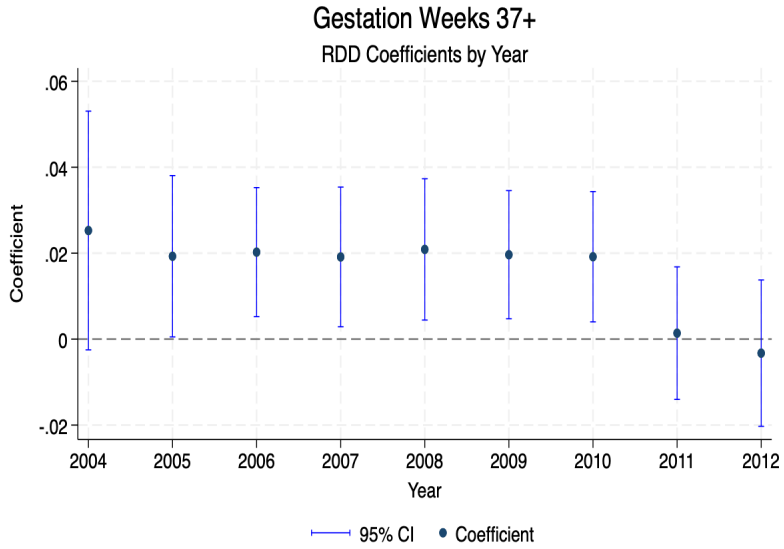
Figure 8



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on low birth weight (<2.5kg) and very low birth weight (<1.5kg) indicators from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

To better understand our results, we investigate how FHP investment affected common causes of low birth weight in newborns. The primary cause is premature birth, commonly defined as any birth that occurs before 37 weeks gestation. In addition to the fact that a baby born early has less time in the mother's uterus to grow and gain weight, much of a fetus's weight is gained during the latter part of the mother's pregnancy. As expected, there was a significant increase in indicator for gestation of at least 37 weeks, as we can see in Figure 9. The results reveal a consistently positive and statistically significant effect from 2004 to 2010, with increases ranging from 1.4 to 1.8 percentage points in full-term births for a 70% increase in FHP transfers. Given control municipalities' full-term birth rates of 93.62% to 94.22%, these increases represent substantial improvements.

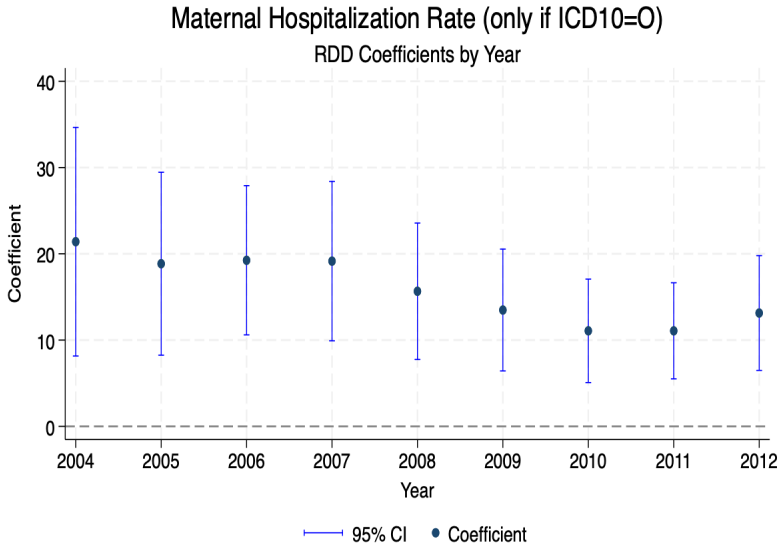
Figure 9



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on gestation of at least 37 weeks indicator from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

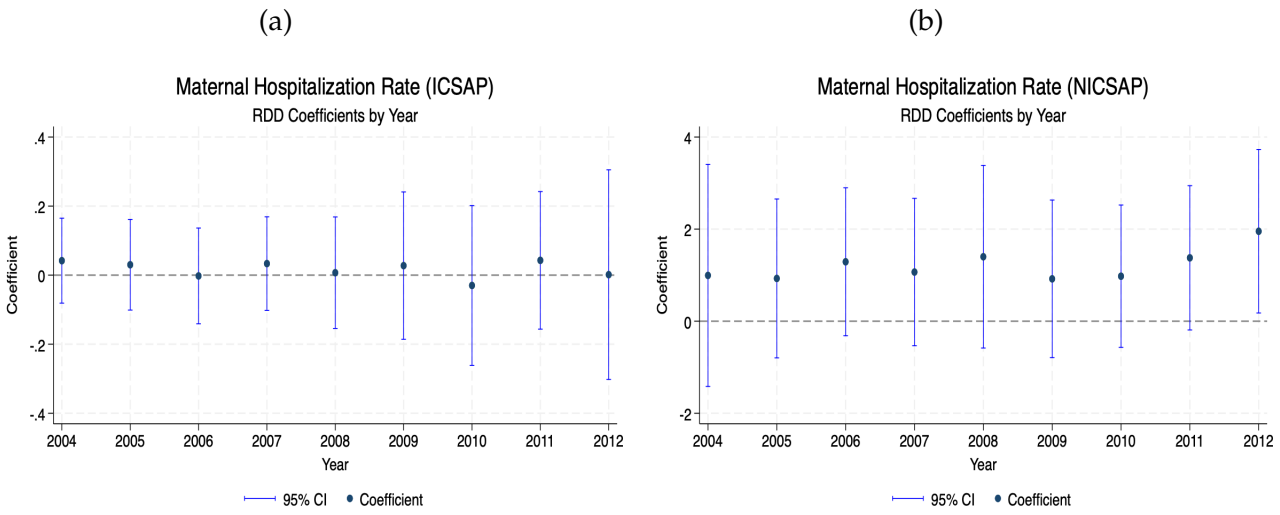
Figure 10 presents the RDD coefficients for Maternal Hospitalization Rate (MHR) from 2004 to 2012, showing the impact of the new FHP funding policy. The results reveal a consistently positive and highly significant effect across all years. We then look into causes of hospitalization that are amenable to primary care (ICSAP), and those that are not (non-ICSAP). Figures 11a and 11b present the treatment effects on Maternal Hospitalization Rates for conditions sensitive to primary care (ICSAP) and not sensitive to primary care (NICSAP). For ICSAP, the results show no statistically significant effects across all years, with coefficients varying around zero. For NICSAP, while most years show positive but insignificant effects, there is a marginally significant increase in 2011 (p=0.085) and a significant increase in 2012 (p=0.031). These findings suggest that while the FHP funding policy did not significantly affect hospitalizations for conditions manageable by primary care, it may have increased access to hospital care for other conditions, particularly in later years. As we show in the previous section, health outcomes of mothers and children indeed improved with FHP's expansion. Therefore, the observed increase in hospitalizations likely reflects enhanced access to healthcare rather than a decline in maternal health. Detailed results are available in Tables 14, 15, and 16 of the appendix.

Figure 10



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on Maternal Hospitalization Rate (MHR) from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

Figure 11



Note: This figure reports the RDD estimates and 95% confidence intervals for the effects of additional FHP transfers on Maternal Hospitalization Rates for conditions sensitive to primary care (ICSAP) and not sensitive to primary care (NICSAP) from 2004 to 2012. The analysis uses the MSE-optimal bandwidth selector for the RD treatment effect estimator, calculated using the method proposed by Calonico, Cattaneo, and Titiunik (2014). The vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, and % households with sanitation, piped water, and sewer.

Despite improvements observed in other health indicators, the analysis of maternal and infant mortality rates from 2004 to 2012 reveals no statistically significant effects of the new FHP funding policy on these critical health outcomes. This lack of significant effect persists across various specifications and alternative definitions of maternal mortality, including total female mortality in reproductive ages (10-49 years). The absence of detectable effects on mortality rates may be attributed to factors such as the rarity of maternal mortality events and the potentially longer exposure periods required for healthcare interventions to impact mortality rates significantly. The limited time window of this study may not capture the full long-term effects of the program on these ultimate health outcomes. Detailed results are available in Tables 17 and 18 in the appendix.

5.3 Robustness

Finally, to address potential violations of the identification strategy, I also consider the possibility of alternative policy discontinuities around the same thresholds of population and/or HDI that could confound the results. To the best of my knowledge, there is no policy discontinuity at either HDI of 0.7 or population of 30,000/50,000 during the period under analysis. Therefore, I investigate alternative discontinuities at neighbour thresholds. The most relevant budgetary component of Brazilian municipalities is the Fundo de Participação dos Municipios (FPM), distributed discontinuously across population brackets. [7] Two of the population thresholds (one at 30,564 and other at 50,940), are near to the FHP population cutoff. Nevertheless, Table 19 shows strong evidence that the FPM is unlikely to drive our results. The absence of a significant effect on health expenditures on the first stage is evidence that the FPM is not generating a funding gap at this threshold. [8]

[7] The thresholds are defined with an associated coefficient that varies nonlinearly between 0.6 and 4, with smaller population brackets corresponding to lower coefficients. Then, each municipality is annually assigned to a population threshold based on its number of inhabitants, with larger municipalities receiving greater transfers.

[8] For future versions of this paper, I plan to conduct additional robustness checks to further strengthen the credibility of the identification strategy. These will include running regressions for the pre-2004 period to establish whether any observed effects were present before the implementation of the policy. I am currently working on the necessary data for this pre-2004 analysis.

6 Conclusion

This paper examines the impact of additional spending on primary health care on maternal and infant health outcomes in the context of Brazil's Family Health Program (FHP). By exploiting discontinuities in FHP funding, I created a quasi-experimental setting to assess how increased resources affected various health indicators from 2004 to 2012.

My findings show that the new FHP funding policy led to significant improvements in healthcare access and key health outcomes. The policy consistently increased FHP coverage, with a 70% rise in annual transfers resulting in a 9 percentage point increase in covered population. It also boosted the number of FHP teams per capita by 0.028 to 0.042 teams per 1000 population and significantly increased home visits, with around 0.4 additional visits per capita. Prenatal care saw substantial enhancements, with an increase of about 6 prenatal visits per 100 women of reproductive age.

Importantly, we observed improvements in several maternal and infant health outcomes. Birth weight consistently increased across all years, with additional gains ranging from 35 to 41 grams for a 70% increase in FHP transfers. While not always statistically significant, there were reductions in the incidence of low birth weight, with the largest reduction being 1.90 percentage points. Full-term births (gestation of at least 37 weeks) increased significantly from 2004 to 2010, with increases ranging from 1.34 to 1.77 percentage points. Maternal hospitalization rates also increased, particularly for conditions not sensitive to primary care, suggesting improved access to necessary hospital care.

However, contrary to our initial hypothesis, we found no significant effects on maternal and infant mortality rates. This lack of impact may be due to the rarity of these events and the limited time window of our study, suggesting that longer exposure periods might be necessary to observe changes in these ultimate health outcomes.

This study makes several important contributions to the literature on primary healthcare interventions in developing countries. First, it provides a unique analysis of the 2004 FHP funding rule change, offering valuable insights into how financial incentives can drive program expansion and improve health outcomes. The finding that even a modest increase in funding led to significant improvements in health outcomes is particularly crucial for resource-constrained settings, demonstrating the potential of targeted investments to yield substantial benefits.

Second, our comprehensive analysis of various outcomes, including birth weight, full-term births, and maternal hospitalizations, offers a more comprehensive view of FHP's impact than previous studies. The observed effects on birth weight (35-41 grams increase) and full-

term births (1.34 to 1.77 percentage points increase) are notably larger than those found in comparable studies, including Bhalotra and Rocha (2019), which found no significant changes in birth weight during the 1996-2004 period. This suggests that the targeted funding increase after 2004 may have enhanced FHP's effectiveness in improving these critical indicators of infant health.

Finally, by examining intermediate outcomes like prenatal visits and home visits, this study sheds light on the pathways through which FHP might improve health outcomes, an aspect not extensively covered in previous literature. The analysis of data over an eight-year period also offers insights into the program's sustained impact over time, addressing a gap in the existing research.

Overall, these results have important policy implications, underscoring the potential of targeted primary healthcare investments to improve maternal and infant health outcomes. While the specific magnitudes of effects should be interpreted cautiously when applied to other contexts, our findings provide strong evidence for the efficacy of programs like the FHP in enhancing healthcare access and outcomes. Future research could benefit from exploring longer-term impacts and investigating the mechanisms through which primary care investments translate into improved health outcomes.

References

Aquino, R., de Oliveira, N. F. and Barreto, M. L. (2009), 'Impact of the family health program on infant mortality in brazilian municipalities', *American journal of public health* **99**(1), 87–93.

Bhalotra, S. R. and Rocha, R. (2019), 'Does universalization of healthwork? evidence from health systems restructuring and expansion in brazil'.

Brazilian Ministry of Health (2006*a*), *Atenção Básica e a Saúde da Família.*, Ministério da Saúde – Brasil, Departamento de Atenção Básica, Brasília (http://dtr2004.saude.gov.br/dab/atencaobasica.php).

Brazilian Ministry of Health (2006*b*), *Saúde da Família no Brasil: Uma Análise de Indicadores Selecionados, 1998 – 2004.*, Série C Projetos, Programas e Relatórios. Brasília: Ministério da Saúde, Brasil.

Calonico, S., Cattaneo, M. D. and Titiunik, R. (2014), 'Robust nonparametric confidence intervals for regression-discontinuity designs', *Econometrica* **82**(6), 2295–2326.

DAB (2000), 'Programa saúde da família – secretaria de políticas de saúde', *Rev Saúde Pública* **34**(3), 316–9.

DATASUS (2023), Ministério da saúde - datasus, Technical report, Ministério da Saúde - DATASUS.

FGV-EPOS (2001), 'Determinação e avaliação do custo do programa de saúde da família', *Brasília, Ministério da Saúde,/REFORSUS* .

Gragnolati, M., Lindelow, M. and Couttolenc, B. (2013), *Twenty years of health system reform in Brazil: an assessment of the Sistema Único de Saúde*, World Bank Publications.

Guanais, F. C. (2013), 'The combined effects of the expansion of primary health care and conditional cash transfers on infant mortality in brazil, 1998-2010.', *American Journal of Public Health* **103**(11).

Harris, M. J. (2014), 'We could all learn from brazil's family health program'.

Jurberg, C. and Humphreys, G. (2010), 'Who brochure', *Bulletin of the World Health Organization* **88**, 641–716.

Macinko, J., Guanais, F. and Marinho De Souza, M. (2006), 'Evaluation of the impact of the family health program on infant mortality in brazil, 1990-2002', *Journal of Epidemiology and Community Health* **60**(1), 13–19.

Macinko, James Almeida, C. and De Sá, P. K. (2007), 'A rapid assessment methodology for the evaluation of primary care organization and performance in brazil', *Health policy and planning* **22**(3), 167–177.

McCrary, J. (2008), 'Manipulation of the running variable in the regression discontinuity design: A density test', *Journal of econometrics* **142**(2), 698–714.

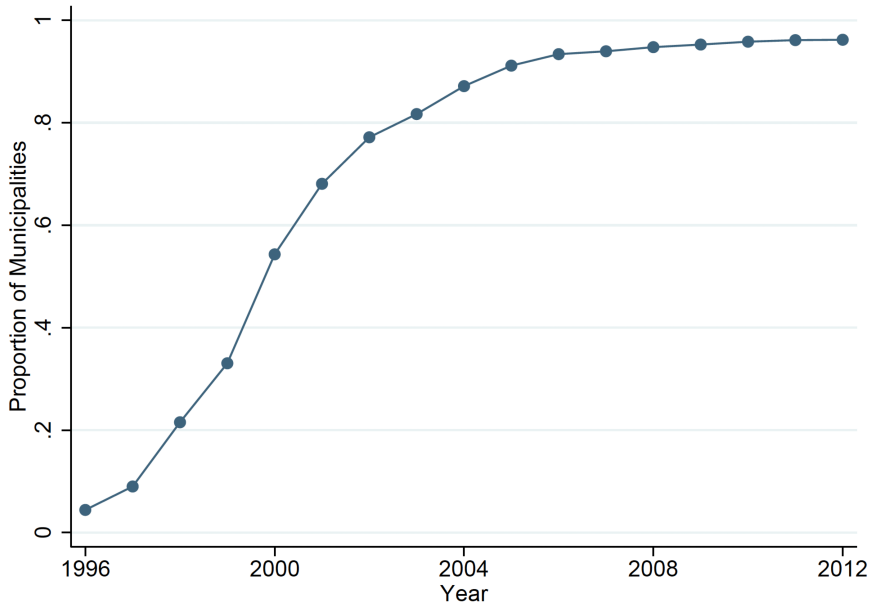
Paim, J., Travassos, C., Almeida, C., Bahia, L. and Macinko, J. (2011), 'The brazilian health system: history, advances, and challenges', *The Lancet* **377**(9779), 1778–1797.

Rocha, R. and Soares, R. R. (2010), 'Evaluating the impact of community-based health interventions: evidence from brazil's family health program', *Health Economics* **19**(S1), 126–158.

A Appendix

A.1 Descriptive Statistics

Figure 12: FHP Expansion



Source: Brazilian Ministry of Health, SAS/Dept de Atenção Básica – DAB.
Note: The original source of data used to compute each variable is listed on Table 2.

Table 1: Main Descriptive Statistics

	Obs.	Mean	Stand Dev	Min	Max	Source of Data	Year
Municipality Public Expenditures and Health Infrastructure							
Dummy for FHP	5560	0.87	0.33	0.00	1.00	Datasus/DAB	2004
Number of FHP teams implemented	5560	3.82	13.34	0.00	619.00	Datasus/DAB	2004
Proportion of FHP estimated covered population	5560	63.02	38.25	0.00	100.00	Datasus/DAB	2004
Total Expenditures, Except in Health (in R$ per capita)	5168	905.35	517.50	34.11	7,564.74	Finbra	2004
Expenditures in Health (in R$ per capita)	5168	293.44	179.47	0.00	2,985.83	Finbra	2004
FHP Expenditures (in R$ per capita)	5560	15.45	11.40	0.00	80.92	FNS	2004
Dummy for Hospital	5560	0.62	0.48	0.00	1.00	Datasus	2004
Hospital Beds (per capita*1000)	5560	1.93	2.44	0.00	46.01	Datasus	2004
Number of Health Facilities with Ambulatory Service							
Total (per capita)	5560	0.50	0.30	0.00	3.22	Datasus/SIA	2004
With Obstetrical / Gyneco. Services (per 1000 women 10-49yo)	5560	0.31	0.41	0.00	3.93	Datasus/SIA	2004
With Pediatric Services (per children 0-1yo)	5560	0.24	0.40	0.00	4.00	Datasus/SIA	2004
Number of Outpatient Procedures							
Total (per capita*1000)	5517	6.81	5.03	0.00	83.34	Datasus/SIA	2004
Number of Gyneco-Obstetrical Appointments (per women*1000 10-49yo)	5517	0.72	0.90	0.00	23.51	Datasus/SIA	2004
Number of Pediatric Appointments (per children*1000 0-1yo)	5517	16.87	19.08	0.00	297.15	Datasus/SIA	2004
Access to Health Services (Mean, Conditional on Birth)							
Birth at Hospital	5560	0.95	0.14	0.00	1.00	Datasus/SINASC	2004
Share C-Sections	5560	0.39	0.18	0.00	1.00	Datasus/SINASC	2004
Prenatal Visits None	5560	0.03	0.05	0.00	0.75	Datasus/SINASC	2004
Prenatal Visits 1-6	5560	0.48	0.24	0.00	0.98	Datasus/SINASC	2004
Prenatal Visits 7+	5560	0.50	0.25	0.00	1.00	Datasus/SINASC	2004

Table 2: Main Descriptive Statistics

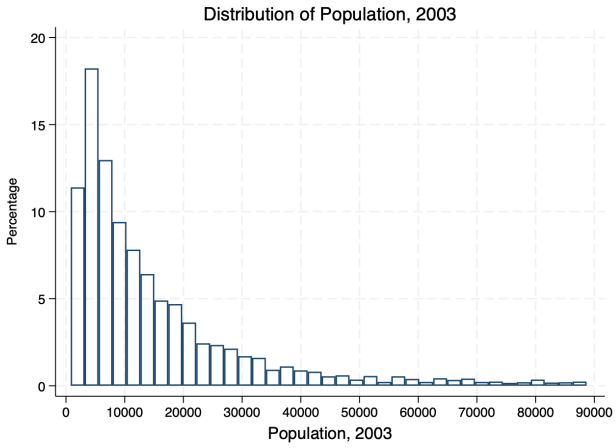
	Obs.	Mean	Stand Dev	Min	Max	Source of Data	Year
Maternal Mortality and Hospitalization Rates (per 1000 women 10-49yo)							
Female Mortality Rate (Irrespective of Cause)	5560	0.97	0.69	0.00	11.05	Datasus/SIM	2004
Maternal Mortality Rate (MMR, only if ICD10="O")	5560	0.03	0.12	0.00	1.93	Datasus/SIM	2004
Female Mortality Rate (Irrespective of Cause, per 1000 Babies 0-1yo)	5560	17.43	13.90	0.00	210.53	Datasus/SIM	2004
Maternal Mortality Rate (MMR, only if ICD10="O", per 1000 Babies 0-1yo)	5560	0.54	2.16	0.00	45.45	Datasus/SIM	2004
Maternal Hospitalization Rate (MHR, only if ICD10="O")	5560	42.72	18.58	0.00	151.05	Datasus/SIH	2004
MHR ICSAP	5560	0.17	0.50	0.00	9.47	Datasus/SIH	2004
MHR Non-ICSAP	5560	8.47	6.60	0.00	94.06	Datasus/SIH	2004
Infant Mortality Rate (IMR, per 1000 babies 0-1yo)							
Total	5560	15.88	15.22	0.00	631.58	Datasus/SIM	2004
Infectious	5560	1.15	4.03	0.00	210.53	Datasus/SIM	2004
Perinatal	5560	8.93	9.72	0.00	210.53	Datasus/SIM	2004
Respiratory	5560	0.99	3.64	0.00	157.89	Datasus/SIM	2004
Congenital	5560	2.16	4.33	0.00	58.82	Datasus/SIM	2004
External	5560	0.29	1.49	0.00	32.26	Datasus/SIM	2004
Others	5560	0.61	2.08	0.00	35.09	Datasus/SIM	2004
Fetal	5560	4.74	6.54	0.00	105.26	Datasus/SIM	2004
Neonatal	5560	10.37	10.59	0	91	Datasus/SIM	2004
Within 24hs	5560	4.11	6.12	0.00	74.07	Datasus/SIM	2004
Within 24hs-27 days	5560	6.31	8.03	0.00	210.53	Datasus/SIM	2004
Within 27 days - 1 year	5560	5.46	9.00	0.00	421.05	Datasus/SIM	2004
Birth Outcomes (Mean, Conditional on Birth)							
Apgar 1	5560	8.12	0.53	2.24	10.00	Datasus/SINASC	2004
Apgar 5	5560	9.27	0.43	2.00	10.00	Datasus/SINASC	2004
Birth Weight	5560	3,209.87	97.40	2,575.50	3,750.14	Datasus/SINASC	2004
Low Birth Weight (<2,5k)	5560	0.07	0.03	0.00	0.40	Datasus/SINASC	2004
Gestation Weeks 37+	5560	0.94	0.05	0.24	1.00	Datasus/SINASC	2004

A.2 Design Validity and Robustness

Table 3: Balancedness Around Cutoffs for the Main Sample

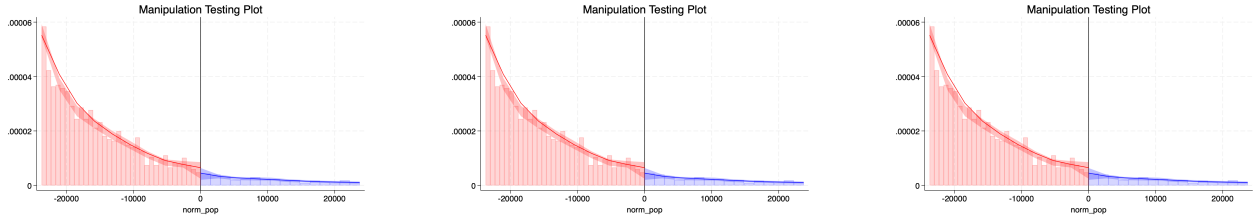
	RD Effect	Robust p-value
Dep Var:		
Area	43.366	.605
Log (Density)	-.067	.605
Log (per capita GDP)	-.258	.781
Total Expenditures, Except in Health (in R$ per capita)	-28.438	.939
Expenditures in Health (in R$ per capita)	-23.075	.781
Households covered by PBF	410.978	.240
Hospital Beds per capita	.014	.686
Hospital Indicator	.044	.635
Households with sewage treatment system	-7.723	.663
Households with garbage collection system	-9.883	.128

Figure 13: Population Histogram



Note: As an initial assessment of how serious the issue of manipulative sorting might be, we plot a population histogram

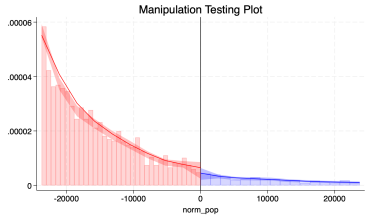
Figure 14: McCrary tests for population thresholds from 2004 to 2012.



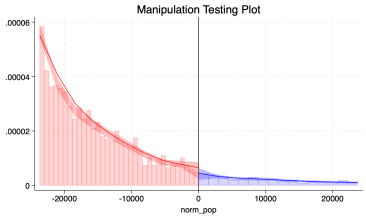
(a) 2004

(b) 2005

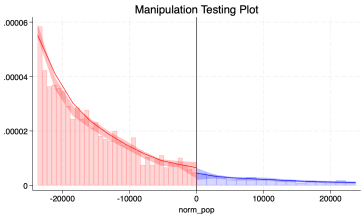
(c) 2006



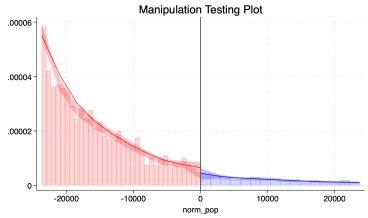
(d) 2007



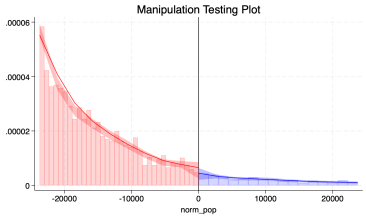
(e) 2008



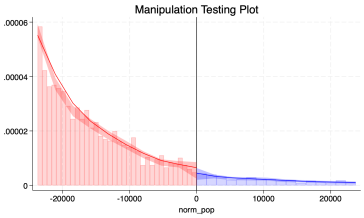
(f) 2009



(g) 2010



(h) 2011

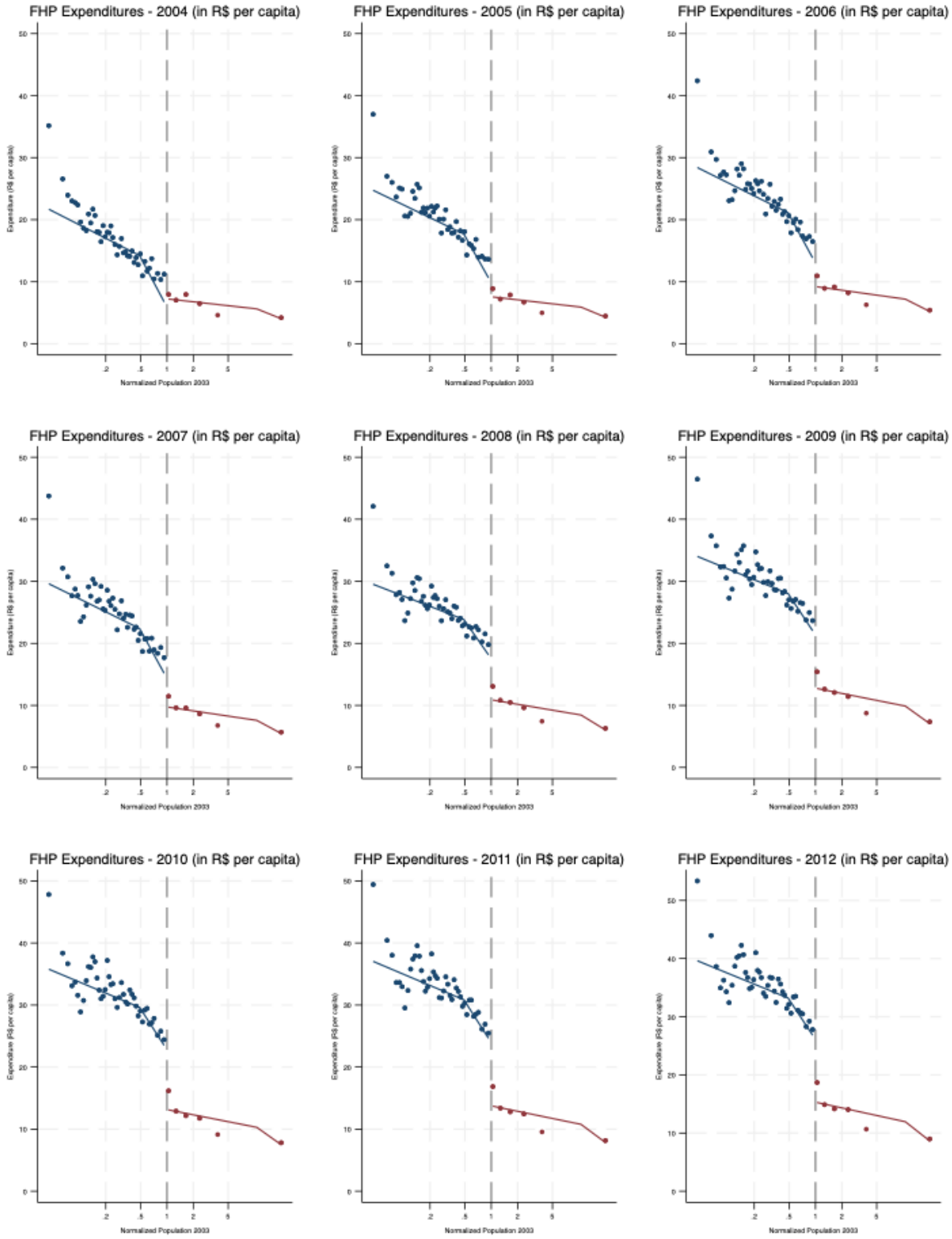


(i) 2012

a

[a] *Note:* The population normalization was performed by subtracting the cutoff value from the population size in 2003, adjusted for specific regions. This test checks for discontinuities in the density of the running variable at the threshold.

Figure 15: FHP Expenditures per capita, 2004-2012



a

[a] *Note:* Each panel shows the relationship between normalized population and FHP expenditures per capita for a given year. The vertical dashed line represents the population cutoff, which is 50,000 for the Norte region, Maranhão (MA), and Mato Grosso (MT), and 30,000 for other areas. The x-axis shows the population normalized by the respective cutoff.

A.3 Results

Table 4: First-Stage Estimates: Population Threshold and FHP Transfers (in log)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	0.4798***	0.5222***	0.7286***	0.7108***	0.7368***	0.7782***	0.7469***	0.7849***	0.6874***
(s.e)	(0.1472)	(0.1351)	(0.1103)	(0.1146)	(0.1094)	(0.1080)	(0.0998)	(0.0924)	(0.0889)
Control Group Mean (in R$)	270,879.19	337,594.52	443,462.57	475,578.87	529,629.11	630,831.92	659,295.77	683,309.62	734,016.87
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the first stage estimates for each year from 2004 to 2012; (ii) FHP Funds are estimated in log(Variable); (iii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iv) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (v) To help evaluate the magnitude of the effects, the dependent variable mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 5: Proportion of FHP Estimated Covered Population

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	6.9543	7.7735	13.3473***	11.9015***	11.6984***	12.1874***	13.0061***	13.8061***	11.4637**
(s.e)	(7.1295)	(5.8880)	(4.0919)	(4.5036)	(4.1966)	(3.9814)	(4.0258)	(3.7479)	(4.4516)
p-value	0.329	0.187	0.001	0.008	0.005	0.002	0.001	0.000	0.010
Control Group Mean	65.122	73.077	76.240	78.323	79.964	81.604	82.943	83.525	83.791
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for the proportion of FHP estimated covered population for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 6: FHP Teams per Capita (*1000)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	0.0195	0.0292	0.0462**	0.0394**	0.0339*	0.0463**	0.0530***	0.0549***	0.0580***
(s.e)	(0.0270)	(0.0212)	(0.0149)	(0.0169)	(0.0183)	(0.0148)	(0.0153)	(0.0149)	(0.0160)
p-value	0.470	0.169	0.002	0.019	0.064	0.002	0.001	0.000	0.000
Control Group Mean	0.147	0.292	0.438	0.585	0.697	0.834	0.989	1.127	1.265
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for FHP Teams per Capita (*1000) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 7: FHP Home Visits per Capita

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	0.3351	0.5205**	0.4927***	0.6370***	0.4246**	0.4432**	0.4313**	0.5333***	0.5410***
(s.e)	(0.2881)	(0.2472)	(0.1887)	(0.2164)	(0.1993)	(0.1759)	(0.2006)	(0.1953)	(0.2082)
p-value	0.245	0.035	0.009	0.003	0.033	0.012	0.032	0.006	0.009
Control Group Mean	2.180	2.385	2.598	2.846	2.633	2.657	2.715	2.769	2.755
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for FHP Home Visits per Capita for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 8: Pregnant Women Followed by FHP Team (per 100 Women in Reproductive Age)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	1.8207	2.0396**	3.4717***	3.5098***	3.5515***	3.5994***	2.7152***	3.4581***	2.3988***
(s.e)	(1.2418)	(1.0147)	(0.9305)	(0.9211)	(0.8912)	(0.7966)	(0.7966)	(0.6522)	(0.6803)
p-value	0.143	0.044	0.000	0.000	0.000	0.000	0.001	0.000	0.000
Control Group Mean	15.837	18.906	18.682	18.495	17.683	17.028	16.691	16.660	16.381
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Pregnant Women Followed by FHP Team (per 100 Women in Reproductive Age) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 9: Prenatal Visits Delivered by FHP Team (per 100 Women in Reproductive Age)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	5.2482***	4.8416***	7.1752***	7.4835***	7.7166***	8.6617***	7.9157***	9.3487***	7.2191***
(s.e)	(1.5074)	(1.4274)	(1.5593)	(1.5392)	(1.6991)	(1.9113)	(1.6918)	(1.7281)	(1.5043)
p-value	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Control Group Mean	17.652	20.052	22.175	23.390	21.749	21.083	20.384	19.991	20.061
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
OObs	4112	4318	4436	4462	4506	4515	4523	4572	4575

Notes: (i) The table reports the treatment effect estimates for Prenatal Visits Delivered by FHP Team (per 100 Women in Reproductive Age) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 10: Birth Weight (in grams)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	57.93***	57.33***	49.96**	57.45***	54.26***	55.31***	51.06***	53.13***	55.19***
(s.e)	(19.31)	(21.64)	(19.70)	(22.05)	(20.52)	(18.97)	(18.88)	(18.18)	(20.53)
p-value	0.003	0.008	0.011	0.009	0.008	0.004	0.007	0.003	0.007
Control Group Mean	3212.36	3218.93	3214.80	3210.24	3209.18	3205.14	3201.35	3196.08	3198.80
Bandwidth	[7731, 7731]	[7907, 7907]	[7730, 7730]	[7731, 7731]	[7137, 7137]	[7225, 7225]	[7169, 7169]	[7974, 7974]	[7557, 7557]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Birth Weight (in grams) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 11: Low Birth Weight Indicator (<2.5kg)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	-0.0020*	-0.0026*	-0.0018	-0.0024**	-0.0017	-0.0014	-0.0016	-0.0013	-0.0014
(s.e)	(0.0012)	(0.0015)	(0.0012)	(0.0012)	(0.0013)	(0.0010)	(0.0013)	(0.0011)	(0.0014)
p-value	0.092	0.081	0.148	0.043	0.196	0.174	0.204	0.234	0.307
Control Group Mean	0.073	0.072	0.074	0.075	0.076	0.076	0.076	0.078	0.078
Bandwidth	[7667, 7667]	[7621, 7621]	[7219, 7219]	[7555, 7555]	[7592, 7592]	[7230, 7230]	[7459, 7459]	[7307, 7307]	[7854, 7854]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Low Birth Weight Indicator (<2.5kg) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 12: Very Low Birth Weight Indicator (<1.5kg)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	-0.0020*	-0.0026	-0.0018	-0.0024**	-0.0017	-0.0014	-0.0016	-0.0013	-0.0014
(s.e)	(0.0012)	(0.0015)	(0.0012)	(0.0012)	(0.0013)	(0.0010)	(0.0013)	(0.0011)	(0.0014)
p-value	0.092	0.081	0.148	0.043	0.196	0.174	0.204	0.234	0.307
Control Group Mean	0.0087	0.0091	0.0104	0.0102	0.0104	0.0112	0.0106	0.0110	0.0112
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	1917	1917	1917	1917	1917	1917	1917	1917	1917

Notes: (i) The table reports the treatment effect estimates for Very Low Birth Weight Indicator (<1.5kg) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 13: Gestation Weeks 37+

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	0.0253*	0.0193**	0.0202***	0.0191**	0.0209**	0.0196**	0.0192**	0.0014	-0.0033
(s.e)	(0.0142)	(0.0096)	(0.0077)	(0.0083)	(0.0084)	(0.0076)	(0.0077)	(0.0079)	(0.0087)
p-value	0.075	0.044	0.008	0.021	0.013	0.010	0.013	0.860	0.707
Control Group Mean	0.9422	0.9417	0.9396	0.9398	0.9387	0.9382	0.9362	0.9038	0.8794
Bandwidth	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]	[7765, 7765]	[7662, 7029]	[7456, 7456]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Gestation Weeks 37+ for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 14: Maternal Hospitalization Rate (MHR, only if ICD10='O')

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	21.403***	18.853***	19.250***	19.153***	15.657***	13.484***	11.073***	11.071***	13.138***
(s.e)	(6.7592)	(5.4095)	(4.4130)	(4.7151)	(4.0349)	(3.6027)	(3.0591)	(2.8408)	(3.3946)
p-value	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Control Group Mean	42.733	42.865	41.832	39.838	36.589	35.988	34.981	34.303	32.953
Bandwidth	[7667, 7667]	[7621, 7621]	[7219, 7219]	[7555, 7555]	[7592, 7592]	[7230, 7230]	[7459, 7459]	[7307, 7307]	[7854, 7854]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Maternal Hospitalization Rate (MHR, only if ICD10='O') for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 15: Maternal Hospitalization Rate (ICSAP)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	0.042	0.030	-0.002	0.033	0.007	0.028	-0.030	0.043	0.001
(s.e)	(0.063)	(0.067)	(0.071)	(0.069)	(0.082)	(0.109)	(0.118)	(0.102)	(0.155)
p-value	0.504	0.653	0.974	0.629	0.932	0.800	0.800	0.673	0.993
Control Group Mean	0.161	0.181	0.176	0.171	0.223	0.258	0.285	0.305	0.337
Bandwidth	[7667, 7667]	[7621, 7621]	[7219, 7219]	[7555, 7555]	[7592, 7592]	[7230, 7230]	[7459, 7459]	[7307, 7307]	[7854, 7854]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Maternal Hospitalization Rate (ICSAP) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 16: Treatment Effects: Maternal Hospitalization Rate (NICSAP)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	0.994	0.928	1.290	1.067	1.400	0.919	0.977	1.376*	1.953**
(s.e)	(1.231)	(0.881)	(0.820)	(0.817)	(1.012)	(0.873)	(0.789)	(0.800)	(0.906)
p-value	0.419	0.292	0.116	0.191	0.167	0.293	0.216	0.085	0.031
Control Group Mean	8.287	8.314	8.427	8.541	9.931	10.094	10.354	10.259	10.443
Bandwidth	[7667, 7667]	[7621, 7621]	[7219, 7219]	[7555, 7555]	[7592, 7592]	[7230, 7230]	[7459, 7459]	[7307, 7307]	[7854, 7854]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Maternal Hospitalization Rate (NICSAP) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 17: Infant Mortality Rate (before 1 year old, per 1000 babies 0-1 year old)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	5.023	3.336	2.526	1.060	2.054	1.482	0.943	1.213	1.408
(s.e)	(2.888)	(3.096)	(2.051)	(1.642)	(1.307)	(1.990)	(2.515)	(1.310)	(2.033)
p-value	0.082	0.281	0.218	0.519	0.116	0.456	0.456	0.354	0.489
Control Group Mean	15.929	14.880	13.838	13.694	13.185	12.696	13.842	13.397	12.881
Bandwidth	[7667, 7667]	[7621, 7621]	[7219, 7219]	[7555, 7555]	[7592, 7592]	[7230, 7230]	[7459, 7459]	[7307, 7307]	[7854, 7854]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Infant Mortality Rate (before 1 year old, per 1000 babies 0-1 year old) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 18: Maternal Mortality Rate (only ICD10="O", per 1000 babies 0-1 year old)

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	0.0093	0.0041	0.0344	0.0213	0.0090	-0.0126	0.0110	0.0206	0.0054
(s.e)	(0.0144)	(0.0133)	(0.0195)	(0.0148)	(0.0110)	(0.0124)	(0.0100)	(0.0113)	(0.0119)
p-value	0.520	0.758	0.078	0.150	0.412	0.309	0.271	0.068	0.647
Control Group Mean	0.0329	0.0334	0.0278	0.0277	0.0312	0.0308	0.0258	0.0271	0.0251
Bandwidth	[7667, 7667]	[7621, 7621]	[7219, 7219]	[7555, 7555]	[7592, 7592]	[7230, 7230]	[7459, 7459]	[7307, 7307]	[7854, 7854]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4154	4358	4474	4505	4542	4567	4595	4610	4617

Notes: (i) The table reports the treatment effect estimates for Maternal Mortality Rate (only ICD10="O", per 1000 babies 0-1 year old) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) To help evaluate the magnitude of the effects, the control group mean — the average for municipalities within a 7000-population interval above the population-threshold — is presented. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.

Table 19: Treatment Effects: Birth Weight - FPM Population Threshold

	Year								
	2004	2005	2006	2007	2008	2009	2010	2011	2012
Coefficient	36.502	43.726	31.168	44.566**	35.650	29.000**	29.709	38.606*	33.827
(s.e)	(23.820)	(44.102)	(25.488)	(21.129)	(25.688)	(14.258)	(19.382)	(22.133)	(32.846)
p-value	0.125	0.321	0.221	0.035	0.165	0.042	0.125	0.081	0.303
Control Group Mean	3212.31	3218.88	3214.75	3210.19	3209.14	3205.09	3201.30	3196.04	3198.78
Bandwidth	[7975, 7975]	[7986, 7986]	[7647, 7647]	[7222, 7422]	[7263, 7263]	[7781, 7781]	[7076, 7076]	[7979, 7979]	[7309, 7309]
Bandwidth Type	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)	MSE-optimal (1)
Kernel	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov	Epanechnikov
Obs	4120	4320	4435	4465	4501	4526	4554	4569	4578

Notes: (i) The table reports the treatment effect estimates for Birth Weight (using FPM population threshold) for each year from 2004 to 2012; (ii) Vector of municipal controls includes PBF coverage, GDP per capita (log), % urban population, % households with sanitation, piped water, and sewer; (iii) Main specification uses the MSE-optimal bandwidth selector for the RD treatment effect estimator calculated by the method proposed by Calonico, Cattaneo, and Titiunik (2014); (iv) Control group mean refers to the average birth weight of babies in municipalities within a 7000-population interval above the population-threshold. *** p<0.01, ** p<0.05, * p<0.1; Robust standard errors in parentheses.